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EXPLICIT BOUNDS FOR THE RIEMANN ZETA-FUNCTION ON THE 1-LINE

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Abstract. Explicit estimates for the Riemann zeta-function on the 1-line are derived using various methods, in particular van der Corput lemmas of high order and a theorem of Borel and Carathéodory.

1. Introduction

Let $\zeta(s)$ denote the Riemann zeta-function. An open problem in analytic number theory is to determine the growth rate of $\zeta(s)$ on the line $s = 1 + it$ as $t \to +\infty$. Assuming the Riemann hypothesis, Littlewood [Lit26, Lit28] showed that the value of $|\zeta(1 + it)|$ is limited to the range

$$\frac{1}{\log\log t} \ll \zeta(1 + it) \ll \log\log t.$$

In comparison, the sharpest estimates requiring no assumption furnish a wider range of the form

$$\frac{1}{(\log t)^{2/3}(\log\log t)^{1/3}} \ll \zeta(1 + it) \ll \log^{2/3} t. \tag{1}$$

The upper bound in the unconditional estimate (1) was made explicit by Ford [For02]. He computed numerical constants c and t_0 such that[1]

$$|\zeta(1 + it)| \le c \log^{2/3} t, \qquad (t \ge t_0).$$

The constants c and t_0 are often too large for standard applications, including explicit bounds on $S(t)$ [Tru14b, HSW22] and explicit zero-density estimates [KLN18]. One instead defaults to an asymptotically worse bound of the form $\zeta(1+it) \ll \log t$ that nevertheless is sharper in a finite range of t of interest.

This motivates our Theorem 1. We establish a new explicit bound on $\zeta(1 + it)$ that is asymptotically sharper than $O(\log t)$ but still good at small values of t.

Theorem 1. *For $t \ge 3$, we have*

$$|\zeta(1 + it)| \le 1.731 \frac{\log t}{\log\log t}.$$

Additionally, using a different method, we prove explicit upper bounds of the same form, on both $1/|\zeta(1 + it)|$ and $|\zeta'(1 + it)/\zeta(1 + it)|$.

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[1] Trudgian [Tru14a] later lowered Ford's value of c from 76.2 to 62.6.

Theorem 2. *For $t \geq 3$, we have*

$$\frac{1}{|\zeta(1+it)|} \leq 431.7 \frac{\log t}{\log\log t}.$$

Furthermore, for $t \geq 500$, if

$$1 \leq \sigma \leq 1 + \frac{9}{250}\frac{\log\log t}{\log t},$$

then

$$\left|\frac{\zeta'}{\zeta}(\sigma+it)\right| \leq 154.5 \frac{\log t}{\log\log t}.$$

Theorem 1 makes explicit a result due to Weyl [Wey21] (see also [Tit86, Section 5.16]), and is sharper than the prior explicit bounds due to Patel [Pat22] on $|\zeta(1+it)|$ for t in the range $\exp(3382) \leq t \leq \exp(3.61 \cdot 10^8)$.[2]

As far as we know, the bounds in Theorem 2 are the first unconditional explicit bounds of their kind. The first assertion of Theorem 2 supersedes [Car+22, Proposition A.2] when $t \geq \exp(\exp(10.07))$, while the second assertion supersedes [Tru15, Table 2 with $(W, R_1) = (12, 40.14)$] for $t \geq \exp(\exp(3.85))$.

Under the assumption of the Riemann Hypothesis, sharper explicit estimates for $\zeta'/\zeta(1+it)$ and $1/\zeta(1+it)$ of the order $\log\log t$ are known [Sim23], [CHS24, Theorem 5], [Chi23]. Much work has also been done on bounding more general L-functions [Lum18], [PS22].

Lastly, it is reasonable to try to adapt the remarkable approaches in [Sou09, p. 984] and [LLS15, Lemma 2.5] to our context. In these articles, very precise bounds, conditional on the Riemann hypothesis, are obtained. One can arrange for these approaches to yield unconditional bounds, and the outcome is similarly constrained by the width of available explicit zeros-free regions. See [Li10], for example, where the method from [Sou09] is used to establish bounds on L-functions for $\sigma \geq 1$.

2. Discussion

One can derive an explicit bound on $|\zeta(1+it)|$ of the form $\log t + \alpha_1$ using a version of the Euler–Maclaurin summation. For example, using the version in [Sim20, Corollary 2], applied with $t_0 = 30\pi$ and $c = 1/4$, together with the elementary harmonic sum bound[3] (see Lemma 8 below),

$$\sum_{n=1}^{N} \frac{1}{n} \leq \log N + \gamma + \frac{1}{2N}, \tag{2}$$

yields after a small numerical computation, $|\zeta(1+it)| \leq \log t - 0.45$ for $t \geq 3$.

Patel [Pat22] replaced the use of the Euler–Maclaurin summation by an explicit version of the Riemann–Siegel formula on the 1-line. Consequently, the size of the main term in the Euler–Maclaurin bound was cut by half. Patel thus obtained $|\zeta(1+it)| \leq \frac{1}{2}\log t + 1.93$ for $t \geq 3$, a substantial improvement at the expense of a larger constant term.

[2] We also note that Theorem 1 is sharper for $t \geq 23256$ than the bound $|\zeta(1+it)| \leq \frac{3}{4}\log t$ given in [Tru14a].

[3] Here, $\gamma = 0.577216\ldots$ is the Euler constant.

In general, we expect that an explicit van der Corput lemma of order k, where $k \geq 2$, will produce a bound of the form

$$|\zeta(1+it)| \leq \frac{1}{k}\log t + \alpha_k, \qquad (t \geq 3). \tag{3}$$

One proceeds by approximating $\zeta(1+it)$ by an Euler–Maclaurin sum $\sum_{n \ll t} n^{-1-it}$, or better yet by a Riemann–Siegel sum $\sum_{n \ll \sqrt{t}} n^{-1-it}$. In either case, roughly speaking, the term $(1/k)\log t$ arises from bounding the subsum with $n \leq t^{1/k}$ via the triangle inequality and the harmonic sum estimate (2), while the constant α_k predominantly arises from bounding the subsum with $n > t^{1/k}$ via van der Corput lemmas of order $\leq k$.

However, the α_k obtained using this approach appear to grow fast with k. For example, in [Pat21], α_5 is about 23-times larger than α_2, which already suggests exponential growth. Despite this, there is still gain in letting k grow with t, provided the growth is slow enough. We let k grow like $\log\log t$, which is permitted by a recent optimized van der Corput test, uniform in k, derived in [Yan24]. Subsequently, we obtain a bound on $|\zeta(1+it)|$ of the form $c_0 \log t/\log\log t$ with an explicit number c_0.

While we could let k increase faster than $\log\log t$, leading to a smaller contribution of the $(1/k)\log t$ term, the α_k thus obtained would likely take over as the main term, asymptotically surpassing $\log t/\log\log t$ in size. This limits the order of the van der Corput lemmas we employ.

In proving Theorem 1, we encounter an unexpected "gap interval", not covered by any of the other explicit estimates, which requires a special treatment. Namely, the interval $I = [\exp(16), \exp(662)]$. To prove the inequality in Theorem 1 for all $t \in I$, we use the following proposition.

Proposition 3. *We have*

$$|\zeta(1+it)| \leq \begin{cases} \dfrac{1}{3}\log t + 4.66, & t \geq \exp(16), \\[2ex] \dfrac{8}{33}\log t + 12.45, & t \geq \exp(88). \end{cases}$$

To clarify the need for Proposition 3, consider that if one inputs the bounds in [Pat22], as below, one finds

$$\max_{t \in I} \frac{\min\left(\log t, \dfrac{1}{2}\log t + 1.93, \dfrac{1}{5}\log t + 44.02\right)}{\log t/\log\log t} \geq 2.539,$$

and the maximum of this ratio occurs around $t_1 = \exp(140.3)$. On the other hand, the savings we otherwise establish here may materialize only when t is well past t_1. Therefore, without the improvement enabled by Proposition 3, we cannot do any better than the bound $|\zeta(1+it)| \leq 2.539\log t/\log\log t$ over $t \in I$. Proposition 3, which is based on a hybrid of van der Corput lemmas of orders $k = 3$, 4 and 5, allows us to circumvent this barrier.

In retrospect, a gap interval like I should be expected. This is because the estimates in [Pat21] rely, essentially, on van der Corput lemmas of orders $k = 2$ and

$k = 5$.[4] In particular, bounds that would follow from using van der Corput lemmas of orders $k = 3$ and $k = 4$ are not utilized, which presumably causes the gap.

3. Proof of Theorem 1

Let t_0 be a positive number to be chosen later, subject to $t_0 \geq \exp(990/7)$. We suppose $t \geq t_0$ throughout this section. (We will use another method for $t < t_0$.)

Theorem 9 supplies a Riemann–Siegel approximation of $\zeta(1+it)$ in terms of two sums, plus a remainder term $\mathcal{R}$. We use the triangle inequality to bound the entire second sum in this theorem. This gives

$$|\zeta(1+it)| \leq \left|\sum_{n=1}^{n_1} \frac{1}{n^{1+it}}\right| + \frac{g(t)}{\sqrt{2\pi}} + \mathcal{R}(t), \qquad n_1 = \lfloor \sqrt{t/(2\pi)} \rfloor. \tag{4}$$

Since $g(t)$ and $\mathcal{R}(t)$ are well-understood, the bulk of the work is in bounding the main sum over $n \leq n_1$.

We introduce two sequences, $\{X_m\}_{0 \leq m \leq M}$ and $\{Y_k\}_{3 \leq k \leq r+1}$. The former sequence X_m will be used to partition the main sum in (4) in a dyadic manner. The latter sequence Y_k will be used to easily bound the partition points X_m from above and below solely in terms of t (or, what is essentially the same, in terms of n_1).

To this end, let h and h_2 be real positive numbers, also to be chosen later, subject to $1 < h \leq 2$ and $h_2 < 1/\log 2$. We use h_2 immediately in defining

$$R := 2^{r-1}, \qquad r := \lfloor h_2 \log\log t \rfloor.$$

Furthermore we will assume that h_2 and t_0 are chosen so that

$$r_0 := \lfloor h_2 \log\log t_0 \rfloor \geq 6. \tag{5}$$

Since $t \geq t_0$ by supposition, $r \geq r_0$. The integer r will correspond to the highest order van der Corput lemma we use in bounding the tail of the main sum in (4).

Furthermore, let

$$K := 2^{k-1}, \qquad \theta_k := \frac{2K}{(k-2)K+4}, \qquad (k \geq 3).$$

In the definition of θ_k, K is determined by k. Also, θ_k decreases monotonically to zero with $k \geq 3$, and is bounded from above by $\theta_3 = 1$.

We now define the previously mentioned sequences $\{X_m\}$ and $\{Y_k\}$. Set

$$X_0 := \frac{1}{\sqrt{2\pi}} t^{1/2}, \qquad X_m := h^{-m} X_0, \qquad (m \geq 1).$$

So, $\lfloor X_0 \rfloor = n_1$ and since $h > 1$, X_m decreases monotonically with m. Also set

$$Y_k := X_0^{\theta_k}, \qquad (k \geq 3), \tag{6}$$

so that Y_k decreases monotonically with $k \geq 3$, starting at $Y_3 = X_0$. Note that if k is large, then $\theta_k \approx 2/k$, and so $Y_k \approx t^{1/k}$ for large k. Lastly, define

$$M := \left\lceil \frac{(1-\theta_{r+1}) \log X_0}{\log h} \right\rceil.$$

[4] The bound obtained via the Riemann–Siegel formula in [Pat21] should correspond to a $k = 2$ van der Corput lemma.

The integer M corresponds to the number of points we will use to partition the main sum. From the definition, we have

$$\frac{(1-\theta_{r+1})\log X_0}{\log h} \leq M < \frac{(1-\theta_{r+1})\log X_0}{\log h} + 1. \tag{7}$$

Thus, our choice of M ensures that

$$X_M \leq X_0^{\theta_{r+1}} = Y_{r+1} < X_{M-1} \leq hX_0^{\theta_{r+1}}. \tag{8}$$

Additionally, we have

$$X_M \leq Y_{r+1} < Y_r < \cdots < Y_4 < Y_3 = X_0. \tag{9}$$

The assumption (5) is required for the subsequent analysis to be valid. This condition implies, among other things, that $\theta_{r+1} < 1$ and, hence, $M \geq 1$. This in turn ensures that some quantities appearing later (e.g. $M-1$) are nonnegative. All contingent conditions we impose (including on r_0, h, h_2 and t_0) will be verified at the end, after choosing the values of our free parameters.

By the chains of inequalities (8) and (9), for all $m \in [1, M-1]$, $Y_{r+1} < X_m < Y_3$. Therefore, since the sequence $Y_{r+1}, \ldots, Y_3$ partitions the interval $[Y_{r+1}, Y_3]$, for each integer $m \in [1, M-1]$, there is a unique integer $k \in [3, r]$ such that

$$X_m \in (Y_{k+1}, Y_k]. \tag{10}$$

Note, though, that multiple m's could correspond to the same k.

With this in mind, let us denote

$$S_m := \sum_{X_m+1<n\leq X_{m-1}} \frac{1}{n^{1+it}} = \sum_{\lfloor X_m\rfloor+1<n\leq \lfloor X_{m-1}\rfloor} \frac{1}{n^{1+it}}, \tag{11}$$

and observe that

$$\left|\sum_{X_{M-1}<n\leq X_0} \frac{1}{n^{1+it}}\right| \leq \sum_{m=1}^{M-1} \left|\frac{1}{(\lfloor X_m\rfloor+1)^{1+it}} + S_m\right| \leq \sum_{m=1}^{M-1} \left(\frac{1}{X_m} + |S_m|\right). \tag{12}$$

We calculate

$$\frac{\lfloor X_{m-1}\rfloor}{\lfloor X_m\rfloor + 1} < \frac{X_{m-1}}{X_m} = h. \tag{13}$$

Here, we used the elementary inequality $x - 1 < \lfloor x\rfloor \leq x$, valid for any real number x together with the definition $X_m = h^{-m}X_0$.

We bound S_m using Lemma 13 with $a = \lfloor X_m\rfloor + 1$ and $b = \lfloor X_{m-1}\rfloor \leq ha$. For each integer $m \in [1, M-1]$ such that $X_m \in (Y_{k+1}, Y_k]$, and with $K = 2^{k-1}$, we have, for any choice of the free parameter $\eta > 0$,

$$\begin{aligned} |S_m| \leq{}& C_k(\eta, h)(\lfloor X_m\rfloor + 1)^{-k/(2K-2)} t^{1/(2K-2)} \\ &+ D_k(\eta, h)(\lfloor X_m\rfloor + 1)^{k/(2K-2)-2/K} t^{-1/(2K-2)}. \end{aligned} \tag{14}$$

To bound the RHS, observe that

$$Y_{k+1} < \lfloor X_m\rfloor + 1 \leq X_m + 1 = X_m\left(1 + \frac{1}{X_m}\right) \leq Y_k\left(1 + \frac{1}{X_m}\right). \tag{15}$$

In addition, $X_m \geq X_{M-1} > X_0^{\theta_{r+1}}$. Moreover, it follows by definition that

$$\theta_{r+1} = \frac{2}{r - 1 + 2^{2-r}}. \tag{16}$$

Since θ_{r+1} decreases continuously with r, and $r \le h_2 \log\log t$, we have $X_0^{\theta_{r+1}} = \exp(\theta_{r+1}\log X_0) \ge \kappa(h_2,t)$ where

$$\kappa(h_2,t) := \exp\left(\frac{\log(t/(2\pi))}{h_2\log\log t - 1 + 2^{2-h_2\log\log t}}\right). \tag{17}$$

It is straightforward to verify that $\kappa(h_2,t)$ is increasing in t for $t \ge t_0$, so $\kappa(h_2,t) \ge \kappa(h_2,t_0)$. Inserting (15) and (17) into (14),

$$\begin{aligned} |S_m| &\le C_k(\eta,h) Y_{k+1}^{-k/(2K-2)} t^{1/(2K-2)} \\ &\quad + \widehat{D}_k(\eta,h,h_2,t_0) Y_k^{k/(2K-2)-2/K} t^{-1/(2K-2)} \end{aligned} \tag{18}$$

for $t \ge t_0$, where

$$\widehat{D}_k(\eta,h,h_2,t_0) := D_k(\eta,h)\left(1+\frac{1}{\kappa(h_2,t_0)}\right)^{k/(2K-2)-2/K}. \tag{19}$$

This is permissible since the exponents of $\lfloor X_m \rfloor + 1$ in the first and second terms of (14) respectively satisfy

$$-\frac{k}{2K-2} < 0, \qquad \text{and} \qquad \frac{k}{2K-2} - \frac{2}{K} \ge 0, \qquad (k \ge 3). \tag{20}$$

It is worth clarifying that the right-hand side of (18) depends on m via the requirement (10).

Next, we express the bound on S_m in (18) solely in terms of t. Starting with the first term, we have

$$Y_{k+1}^{-k/(2K-2)} t^{1/(2K-2)} = (2\pi)^{\theta_{k+1}k/(4K-4)} t^{-\rho(k)}, \tag{21}$$

where after a quick calculation,

$$\rho(k) := \frac{K-2}{K-1}\cdot\frac{1}{2(k-1)K+4}. \tag{22}$$

We estimate the second term in (18) similarly, yielding

$$\begin{aligned} Y_k^{k/(2K-2)-2/K} t^{-1/(2K-2)} &< (2\pi)^{-\theta_k(k/(4K-4)-1/K)} t^{-\frac{K}{K-1}\cdot\frac{1}{2(k-2)K+4}} \\ &< (2\pi)^{-\theta_k(k/(4K-4)-1/K)} t^{-\rho(k)}. \end{aligned} \tag{23}$$

The exponent of 2π in the estimate (21) may be bounded for $k > 5$ via $\theta_{k+1} \le 32/81$. Additionally, recalling the second inequality of (20), we may bound the contribution of the 2π-factor in the estimate (23) simply by 1. This motivates us to define

$$A(k) := \begin{cases} (2\pi)^{\theta_{k+1}\frac{k}{4(K-1)}} C_k + (2\pi)^{-\theta_k(\frac{k}{4(K-1)}-\frac{1}{K})}\widehat{D}_k, & 3 \le k \le 5, \\ (2\pi)^{\frac{8k}{81(K-1)}} C_k + \widehat{D}_k, & 6 \le k. \end{cases} \tag{24}$$

Note that $A(k)$ additionally depends on η, h, h_2 and t_0. Assembling (18), (21), (23) and (24), we obtain that if $1 \le m \le M-1$ and k is determined via the requirement (10), then

$$|S_m| < A(k)\, t^{-\rho(k)}. \tag{25}$$

We now divide the argument into two cases: $3 \le k \le 5$ and $k > 5$. In the former case, let N_k denote the number of sums S_m corresponding to each integer $k \ge 3$. In

other words, N_k is the number of m such that $X_m \in (Y_{k+1}, Y_k]$. We observe that $X_m \in (Y_{k+1}, Y_k]$ if and only if $M_k \le m < M_{k+1}$, where

$$M_k := \frac{(1-\theta_k)\log X_0}{\log h}. \tag{26}$$

Therefore, by the bound on $|S_m|$ in (25),

$$\sum_{1\le m<M_6} |S_m| < \sum_{k=3}^{5} N_k\, A(k)\, t^{-\rho(k)}. \tag{27}$$

By counting the maximum number of integers in a continuous interval, for $k \ge 3$,

$$N_k \le \frac{(\theta_k - \theta_{k+1})\log X_0}{\log h} + 1 \le \phi_k \log t, \tag{28}$$

where

$$\phi_k = \phi_k(h, t_0) := \frac{\theta_k - \theta_{k+1}}{2\log h} + \frac{1}{\log t_0}. \tag{29}$$

Thus, in view of (27) we obtain

$$\sum_{1\le m<M_6} |S_m| \le b_0, \tag{30}$$

where by (28) and on explicitly computing the values of $\rho(k)$ for $3 \le k \le 5$,

$$b_0 = b_0(\eta, h, h_2, t_0) := A(3)\phi_3 \frac{\log t_0}{t_0^{1/30}} + A(4)\phi_4 \frac{\log t_0}{t_0^{3/182}} + A(5)\phi_5 \frac{\log t_0}{t_0^{7/990}}. \tag{31}$$

Here, we used the supposition $t \ge t_0 \ge \exp(990/7)$ which ensures that $\log t/t^{\rho(k)}$ is decreasing for $t \ge t_0$, with $3 \le k \le 5$.

Next, for $k > 5$, we use a tail argument that removes the dependence of A on k in the bound (25). Define

$$A_0 = A_0(\eta, h, h_2, t_0) := \sup_{k\ge 6} A(k). \tag{32}$$

We will bound A_0 explicitly later (Section 4), after determining η, h, h_2 and t_0. Importantly, under our assumptions, it will transpire that A_0 is finite.

In the meantime, we observe that $\rho(k)$ is continuously decreasing in $k \ge 3$ (towards zero), as can be seen from computing the derivative of $\rho(k)$ with respect to k. So, by simply counting the number of terms, and noting that as m ranges over $[M_6, M-1]$, $k \le r$, we obtain from (25),

$$\sum_{M_6\le m\le M-1} |S_m| \le |M - M_6| A_0\, t^{-\rho(r)} \le A_1\, \frac{|b_1 - \theta_{r+1}|}{t^{\rho(r)}} \log t, \tag{33}$$

where, from (7) and (26),

$$A_1 = A_1(\eta, h, h_2, t_0) := \frac{1}{2}\cdot\frac{A_0}{\log h}, \qquad b_1 = b_1(h, t_0) := \theta_6 + \frac{2\log h}{\log t_0}.$$

We want to further simplify the bound on the right-side of (33). Via (16) and a routine calculation,

$$\frac{2}{r - 1 + 2^{2-r_0}} \le \theta_{r+1} < b_1, \qquad (r \ge r_0 \ge 6). \tag{34}$$

Hence, utilizing the inequality $r \le h_2 \log\log t$ as well as $\log(u) \le u-1$, valid for any real number $u > 0$, yields that if $r \ge 6$, then

$$\frac{b_1 - \theta_{r+1}}{t^{\rho(r)}} \le \exp(b_1 - \theta_{r+1} - 1 - \rho(r)\log t) \le \frac{1}{A_2}\frac{1}{\log t}, \tag{35}$$

where (identifying x with $\log t$ below)

(36)

$$A_2 = A_2(h, h_2, t_0) := \inf_{x \ge \log t_0} \frac{1}{x} \exp\left(1 - b_1 + \frac{2}{h_2 \log x - 1 + 2^{2-r_0}} + x\,\rho(h_2 \log x)\right).$$

In indicating the dependencies of A_2, we used that r_0 is determined by h_2 and t_0, and b_1 further depends on h. Now, from the definition of ρ, and using

$$2^{h_2 \log x} = x^{h_2 \log 2},$$

we obtain[5]

$$\rho(h_2 \log x) = \frac{x^{h_2 \log 2} - 4}{x^{h_2 \log 2} - 2} \cdot \frac{1}{(h_2 \log x - 1)x^{h_2 \log 2} + 4}. \tag{37}$$

Since[6] by supposition $0 < h_2 < 1/\log 2$, the quantity defining A_2 tends to $+\infty$ as $x \to +\infty$. As this quantity is continuous and positive for $x \ge \log t_0$, A_2 is finite and positive for all admissible h_2. From (33) and (35) we conclude that

$$\sum_{M_6 \le m \le M-1} |S_m| \le \frac{A_1}{A_2}. \tag{38}$$

Lastly, to bound the remaining terms in (12), in view of (7), we have the rough estimate

$$\sum_{m=1}^{M-1} \frac{1}{X_m} = \frac{1}{X_0} \sum_{m=1}^{M-1} h^m = \frac{1}{X_0}\frac{h^M - h}{h-1} < \frac{X_0^{1-\theta_{r+1}} - 1}{X_0}\frac{h}{h-1} < \frac{h}{h-1}\frac{1}{X_0^{\theta_{r+1}}}.$$

Since $r \le h_2 \log\log t$ and $\theta_{r+1} > 2/r$, we have, for $t \ge t_0$,

$$\sum_{m=1}^{M-1} \frac{1}{X_m} < \frac{h}{h-1} \exp\left(-\frac{\log(t_0/(2\pi))}{h_2 \log\log t_0}\right) =: E(h, h_2, t_0), \tag{39}$$

say.

In summary, combining (12), (30), (38) and (39), the tail sum satisfies the inequality

$$\left|\sum_{X_{M-1} < n \le X_0} \frac{1}{n^{1+it}}\right| \le \frac{A_1(\eta, h, h_2, t_0)}{A_2(h, h_2, t_0)} + b_0(\eta, h, h_2, t_0) + E(h, h_2, t_0). \tag{40}$$

So, for such r, the tail sum is bounded by a constant independent of t.

It remains to bound the initial sum over $1 \le n \le X_{M-1}$. We use the triangle inequality followed by the harmonic sum bound (2), which yields

$$\left|\sum_{1 \le n \le X_{M-1}} \frac{1}{n^{1+it}}\right| \le \log X_{M-1} + \gamma + \frac{1}{2(X_{M-1} - 1)}. \tag{41}$$

[5]By the lower bound on h_2 in (5), we have $x^{h_2 \log 2} \ge 64$ for $x \ge \log t_0$. So, $\rho(h_2 \log x)$ is well-defined for $x \ge \log t_0$.

[6]This step is where a slow enough growth on r is utilized, to ensure that $(\log t)^{h_2 \log 2} = o(\log t)$ as $t \to \infty$, so that the constant $A_2(h_2, t_0)$ defined in (36) is positive.

We use the inequality (8) to bound X_{M-1} from above and below, so that the right-side of (41) satisfies

$$\leq \frac{\theta_{r+1}}{2}\log\frac{t}{2\pi} + \log h + \gamma + \frac{1}{2(X_0^{\theta_{r+1}}-1)}. \tag{42}$$

Since $\theta_{r+1} < 2/(r-1)$ by (16), the first term in (42) is bounded by

$$\frac{\log(t/(2\pi))}{r-1} < \frac{\log t - \log 2\pi}{h_2 \log\log t - 2} \leq \frac{1-\frac{\log 2\pi}{\log t_0}}{h_2 - \frac{2}{\log\log t_0}}\frac{\log t}{\log\log t}, \tag{43}$$

where the last inequality is due to the fact that $(1-\frac{\log 2\pi}{x})/(h_2-\frac{2}{\log x})$ is decreasing for $x \geq 990/7$ if $h_2 < 1/\log 2$.

In addition, we use (17) to bound the last term. Putting this together with (41), (42) and (43), we obtain

$$\left|\sum_{1\leq n\leq X_{M-1}} \frac{1}{n^{1+it}}\right| < A_3\frac{\log t}{\log\log t}, \tag{44}$$

where

$$A_3 = A_3(h,h_2,t_0) := \frac{1-\frac{\log 2\pi}{\log t_0}}{h_2-\frac{2}{\log\log t_0}} + \left(\log h + \gamma + \frac{1}{2(\kappa(h_2,t_0)-1)}\right)\frac{\log\log t_0}{\log t_0}.$$

Lastly, the bound on $\zeta(1+it)$ in (4) carries an extra term $g(t)/\sqrt{2\pi}+\mathcal{R}(t)$. Using the definitions in Theorem 9, this term is easily bounded by $g(t_0)/\sqrt{2\pi}+\mathcal{R}(t_0)$. Therefore, combining this with (40) and (44), we see that

$$|\zeta(1+it)| \leq \left|\sum_{1\leq n\leq n_1}\frac{1}{n^{1+it}}\right| + \frac{g(t_0)}{\sqrt{2\pi}} + \mathcal{R}(t_0) \leq A_4\frac{\log t}{\log\log t},$$

for $t \geq t_0$, where

$$A_4 = A_4(\eta,h,h_2,t_0) := A_3 + \left(\frac{A_1}{A_2} + b_0 + E + \frac{g(t_0)}{\sqrt{2\pi}} + \mathcal{R}(t_0)\right)\frac{\log\log t_0}{\log t_0}.$$

We choose the following values for our free parameters, which are suggested by numerical experimentation.

$$\eta = 1.2364, \qquad h = 1.0224, \qquad h_2 = 0.85532, \qquad t_0 = \exp(3381). \tag{45}$$

With these choices, our preconditions on t_0 and h_2 hold, and the conditions $h > 1$ and $r_0 \geq 6$ also hold (in fact, $r_0 = 6$). Additionally, in Section 4, we show that

$$A_0 = 0.0976483443\ldots,$$
$$A_2 = 0.0618341521\ldots.$$

Therefore, we can explicitly compute

$$b_0 \leq 4.629\cdot 10^{-8},$$
$$A_1 \leq 2.2039724975\ldots,$$
$$A_3 \leq 1.6420608139\ldots,$$
$$A_4 \leq 1.7301296329\ldots.$$

In summary, the assertion of Theorem 1 holds when $t \geq t_0 = \exp(3381)$.

For smaller $t \le t_0$ we use the bound in Patel [Pat22] and Proposition 3. Specifically, the bound in [Pat22, Theorem 1.1] yields

$$|\zeta(1+it)| \le \frac{1}{5}\log t + 44.02 \le 1.731\frac{\log t}{\log\log t}, \qquad \exp(662) \le t \le \exp(3381),$$
$$|\zeta(1+it)| \le \frac{1}{2}\log t + 1.93 \le 1.731\frac{\log t}{\log\log t}, \qquad 3 \le t \le \exp(16).$$

This leaves the gap interval $\exp(16) \le t \le \exp(662)$ for which we invoke Proposition 3. Since the gap interval falls in the range of t where Proposition 3 is applicable, our target bound is verified there as well.

4. Bounding A_0 and A_2

We derive an upper bound on $A_0(\eta, h, h_2, t_0)$, where the parameter values are given in (45). For the convenience of the reader, we recall the definition $A_0 := \sup_{k\ge 6} A(k)$, where for $k \ge 6$,

$$A(k) := (2\pi)^{k/(81K-81)} C_k(\eta, h) + \widehat{D}_k(\eta, h, h_2, t_0).$$

With our choice of free parameter values in (45), we numerically compute the following list of $A(k)$.

$$A(6) = 0.0976483443\ldots, \qquad A(7) = 0.0914388403\ldots,$$
$$A(8) = 0.0882612019\ldots, \qquad A(9) = 0.0865130096\ldots,$$
$$A(10) = 0.0855306439\ldots.$$

The values of C_k and D_k required for the above computation are obtained using the formulas in Lemma 13. In turn, evaluating $C_k(\eta, h)$ and $D_k(\eta, h)$ relies on evaluating $A_k(\eta, h^k)$ and $B_k(\eta)$. When $k = 3$, A_k and B_k are computed directly using their definitions in Lemma 11. When $k \ge 4$, the recursive formulas in Lemma 12 are used to compute them. Let us examine these recursive formulas.

First, the following quantity, which appears in the formula for $A_{k+1}(\eta, \omega)$ in (92), satisfies

$$\max_{k\ge 10} \frac{2^{19/12}(K-1)}{\sqrt{(2K-1)(4K-3)}} = \frac{2^{19/12}\cdot 511}{\sqrt{2092035}} =: \mu_{10}. \tag{46}$$

Here, $K = 2^{k-1}$ and the maximum occurs when $k = 10$. Another quanitity appearing in the formula for $A_{k+1}(\eta, \omega)$ is δ_k, which is clearly seen to decrease with k. Noting that $h > 1$, hence $\omega = h^{k+1} > 1$ for any $k \ge 0$, we deduce that

$$A_{k+1}(\eta, \omega) \le \delta_{10}\left(1 + \mu_{10} A_k(\eta, \omega)^{1/2}\right).$$

So, we are led to consider the discrete map

$$x_{n+1} = \delta_{10}\left(1 + \mu_{10} x_n^{1/2}\right).$$

By a routine calculation, this map has a single fixed point at

$$x^* = \left(\frac{\mu_{10}\delta_{10}}{2} + \sqrt{\frac{\mu_{10}^2\delta_{10}^2}{4} + \delta_{10}}\right)^2 = 2.7600429449\ldots, \tag{47}$$

which is a stable point. Since $A_{k+1}(\eta,\omega) \le x_{k+1}$ for $k \ge 10$ and, by a direct numerical computation, $A_{10}(\eta,\omega) \le x^*$, it follows that

$$A_k(\eta, h^k) \le x^*, \qquad (k \ge 10). \tag{48}$$

The analysis of B_{k+1} for $k \ge 10$ proceeds similarly. The recursive formula for B_{k+1} contains the following quantity satisfying

$$\max_{k\ge 10} \frac{2^{3/2}(K-1)}{\sqrt{(2K-3)(4K-5)}} \le \frac{2^{3/2}\cdot 511}{3\sqrt{231767}} =: \nu_{10}. \tag{49}$$

So, $B_{k+1} \le \delta_{10}\nu_{10}B_k^{1/2}$ for $k \ge 10$. Consider the discrete map $y_{n+1} = \delta_{10}\nu_{10}y_n^{1/2}$. This map has a single fixed point, namely,

$$y^* = (\delta_{10}\nu_{10})^2 = 1.0023463404\ldots,$$

which is also stable. Since $B_{k+1} \le y_{k+1}$ for $k \ge 10$ and, by a direct numerical computation, $B_{10} \le y^*$, it follows that

$$B_k(\eta) \le y^*, \qquad (k \ge 10). \tag{50}$$

Furthermore, we numerically verify that for $k \ge 10$,

$$h^{2k/K-k/(2K-2)}(h-1)\left(\frac{(k-1)!}{2\pi}\right)^{\frac{1}{2K-2}} < 0.023,$$

$$h^{k/(2K-2)}(h-1)^{1-2/K}\left(\frac{2\pi}{(k-1)!}\right)^{\frac{1}{2K-2}} < 0.023.$$

Combining this with (48) and (50), and recalling the definitions in Lemma 13 and (19), yields the inequalities $C_k(\eta,h) < 0.023x^*$ and $\widehat{D}_k(\eta,h) < 0.024y^*$ for $k \ge 10$. Consequently, $A(k) < 0.09$ for $k \ge 10$.

Furthermore, in view of the values of $A(k)$ listed at the beginning, we also see that $A(k) \le A(6) = 0.09764\ldots$ for $6 \le k \le 10$. We therefore conclude $A_0 = A(6)$.

For the computation related to A_2, we shall use a derivative calculation to show that the function

$$a_2(u) := 1 - b_1 + e^u\,\rho(h_2u) + \frac{2}{h_2u - 1 + 2^{2-r_0}} - u \tag{51}$$

is increasing in $u \ge u_0 := \log\log t_0$. So, on identifying u with $\log x$ in the definition of A_2 in (36), the value of A_2 is $\exp(a_2(u_0))$.

To this end, we first calculate

$$\frac{d}{du}e^u\,\rho(h_2u) = e^u\rho(h_2u)\left[1 + \frac{d}{du}\log\rho(h_2u)\right], \qquad \frac{d}{du}\log\rho(h_2u) = -h_2\tilde{a}_2(h_2u),$$

where

$$\tilde{a}_2(v) = \frac{(v-1)\log 2 + 1 - 2^{-v}(8(v-1)\log 2 + 6) + 4^{-v}(8(v-2)\log 2 + 8)}{(v - 1 + 2^{2-v})(1-2^{1-v})(1-2^{2-v})}.$$

Note that v is identified with h_2u, so we need only consider $v \ge h_2u_0 =: v_0$. It is easy to see that

$$\sup_{v\ge v_0}\tilde{a}_2(v) < \sup_{v\ge v_0}\frac{(v-1)\log 2 + 1}{(v - 1 + 2^{2-v_0})(1-2^{1-v_0})(1-2^{2-v_0})} \le \log 2 + 0.2,$$

where the supremum occurs at $v = v_0$. Hence, the derivative of $e^u\,\rho(h_2u)$ is at least $e^u\rho(h_2u)(1-(\log 2+0.2)h_2)$ for $u \ge u_0$. Importantly, $1-(\log 2+0.2)h_2$ is positive. In comparison, the derivative of the remaining terms in the formula for $a_2(u)$ is

$$\frac{d}{du}\left(\frac{2}{h_2u-1+2^{2-r_0}}-u\right) = -\frac{2h_2}{(h_2u-1+2^{2-r_0})^2}-1 \ge -1.04731\ldots,$$

since $u \ge u_0$ and $r_0 = 6$. Since

$$e^u\rho(h_2u)(1-(\log 2+0.2)h_2) \ge 1.06112\ldots \tag{52}$$

for $u \ge u_0$, the derivative of $a_2(u)$ is positive for $u \ge u_0$, proving our claim.

5. Proof of Proposition 3

The calculation is similar to that in [Hia16] on the 1/2-line, so we shall be brief in some details. We divide the main sum in (4) into pieces of length $\approx t^{1/3}$, then apply the van der Corput Lemma 10 to each piece other than an initial segment, which is bounded by the triangle inequality.

Let t_1 be a positive number to be chosen later, subject to $t_1 \ge \exp(12)$. Suppose that $t \ge t_1$, unless otherwise stated, and let n_1 be as defined in (4). Let $P = \lceil t^{1/3} \rceil$, $Q = \lfloor n_1/P \rfloor$, and let q_0 a positive integer to be chosen later, subject to $q_0 \le Q$. Thus, we have

$$\left|\sum_{n=1}^{n_1}\frac{1}{n^{1+it}}\right| \le \sum_{n=1}^{q_0P-1}\frac{1}{n} + \sum_{q=q_0}^{Q-1}\Big|\sum_{n=qP}^{(q+1)P-1}\frac{1}{n^{1+it}}\Big| + \Big|\sum_{QP}^{n_1}\frac{1}{n^{1+it}}\Big|. \tag{53}$$

The harmonic sum bound (2) yields, subject to $q_0P > 1$,

$$\sum_{n=1}^{q_0P-1}\frac{1}{n} \le \log(q_0P-1)+\gamma+\frac{1}{2(q_0P-1)}. \tag{54}$$

For $q_0 \le q \le Q$, let $N = qP$ and let N' be an integer such that $N \le N' < N+P$. Partial summation gives

$$\left|\sum_{n=N}^{N'}\frac{1}{n^{1+it}}\right| \le \frac{1}{N}\max_{1\le L\le P}|S_N(L)|, \tag{55}$$

where

$$S_N(L) := \sum_{n=0}^{L-1}\exp(2\pi i f(n)) \qquad \text{and} \qquad f(x) = \frac{t}{2\pi}\log(N+x).$$

We estimate $S_N(L)$ using Lemma 10 with $r = q$, $K = P$ and

$$K_0 = t_1^{1/3} \le P.$$

This gives, for any positive η,

$$\frac{1}{N}\max_{1\le L\le P}|S_N(L)| \le \frac{1}{q}\sqrt{\frac{\alpha}{W^{1/3}}+\frac{\beta W^{1/3}}{P}+\frac{\alpha\eta}{P}+\frac{\eta\beta W^{2/3}}{P^2}}.$$

where W, λ, $\alpha = \alpha(W,\lambda)$, $\beta = \beta(W)$ are defined in Lemma 10. We estimate the (nonnegative) terms under the square-root as follows.

$$\frac{\alpha}{W^{1/3}} < \frac{\alpha}{\pi^{1/3}q}, \qquad \frac{\beta W^{1/3}}{P} \le \frac{\beta\pi^{1/3}(1+1/q_0)q}{t^{1/3}},$$
$$\frac{\alpha\eta}{P} \le \frac{\alpha\eta}{t^{1/3}}, \qquad \frac{\eta\beta W^{2/3}}{P^2} \le \frac{\eta\beta\pi^{2/3}(1+1/q_0)^2q^2}{t^{2/3}}.$$

Isolating the first term using the inequality $\sqrt{x+y} \le \sqrt{x}+\sqrt{y}$, valid for nonnegative x and y, and using the inequality $q \le Q \le t^{1/6}/(2\pi)^{1/2}$ to bound q in the remaining terms, therefore gives

$$\frac{1}{N}\max_{1\le L\le P}|S_N(L)| \le \frac{\alpha(W,\lambda)^{1/2}}{\pi^{1/6}q^{3/2}} + \frac{B(W,\lambda)}{qt^{1/12}}, \tag{56}$$

where, since $t \ge t_1$,

$$B(W,\lambda) := \sqrt{\frac{1+1/q_0}{2^{1/2}\pi^{1/6}}\beta(W) + \frac{\eta}{t_1^{1/6}}\alpha(W,\lambda) + \frac{\eta(1+1/q_0)^2}{2\pi^{1/3}t_1^{1/6}}\beta(W)}.$$

We also note that for all $q \ge q_0$ we have $W \ge W_0$ and $\lambda \le \lambda_0$, where

$$W_0 := \pi(q_0+1)^3, \qquad \lambda_0 := \left(1+\frac{1}{q_0}\right)^3.$$

Therefore, for all q under consideration, $\alpha(W) \le \alpha(W_0,\lambda_0)$ and $\beta(W) \le \beta(W_0)$. It follows that for $q \ge q_0$,

$$\frac{1}{N}\max_{1\le L\le P}|S_N(L)| \le \frac{\alpha_0^{1/2}}{\pi^{1/6}q^{3/2}} + \frac{B_0}{qt^{1/12}}. \tag{57}$$

where $\alpha_0 := \alpha(W_0,\lambda_0)$ and $B_0 := B(W_0,\lambda_0)$.

Put together, we have

$$\left|\sum_{n=q_0P}^{n_1}\frac{1}{n^{1+it}}\right| \le \frac{\alpha_0^{1/2}}{\pi^{1/6}}\sum_{q=q_0}^{Q}\frac{1}{q^{3/2}} + \frac{B_0}{t^{1/12}}\sum_{q=q_0}^{Q}\frac{1}{q}. \tag{58}$$

Moreover, from the argument in [HPY24, (3.1)], and using the bound $Q \le t^{1/6}/(2\pi)^{1/2}$ once again,

$$\begin{aligned}\sum_{q=q_0}^{Q}\frac{1}{q^{3/2}} &\le \int_{q_0-1/2}^{Q+1/2}\frac{\mathrm{d}x}{x^{3/2}} < \frac{2}{\sqrt{q_0-1/2}},\\ \sum_{q=q_0}^{Q}\frac{1}{q} &\le \int_{q_0-1/2}^{Q+1/2}\frac{\mathrm{d}x}{x} < \frac{1}{6}\log t + \log\tau_1 - \log(q_0-1/2),\end{aligned} \tag{59}$$

where

$$\tau_1 := \frac{1}{(2\pi)^{1/2}} + \frac{1}{2t_1^{1/6}}.$$

Therefore, observing that $\frac{1}{6}\frac{\log t}{t^{1/12}}$ is decreasing for $t \geq t_1 \geq \exp(12)$, and combining this with the initial sum bound (54), we obtain

$$\left|\sum_{n=1}^{n_1} \frac{1}{n^{1+it}}\right| < \log(q_0P-1) + \gamma + \frac{1}{2(q_0P-1)} + \frac{\alpha_0^{1/2}}{\pi^{1/6}} \frac{2}{\sqrt{q_0-1/2}} + \frac{B_0}{t_1^{1/12}} \left(\frac{1}{6}\log t_1 + \log \tau_1 - \log(q_0-1/2)\right).$$

We now estimate

$$\log(q_0P-1) \leq \log(q_0t^{1/3}+q_0-1) \leq \frac{1}{3}\log t + \log\left(q_0 + (q_0-1)t_1^{-1/3}\right),$$

which follows using $P < t^{1/3}+1$. Furthermore,

$$\frac{1}{2(q_0P-1)} \leq \frac{1}{2\left(q_0t_1^{1/3}-1\right)}.$$

This yields

$$|\zeta(1+it)| \leq \frac{1}{3}\log t + B_1(q_0) + \frac{g(t_1)}{\sqrt{2\pi}} + \mathcal{R}(t_1), \tag{60}$$

where

$$B_1(q_0) := \gamma + \log\left(q_0 + (q_0-1)t_1^{-1/3}\right) + \frac{1}{2\left(q_0t_1^{1/3}-1\right)} + \frac{\alpha_0^{1/2}}{\pi^{1/6}} \frac{2}{\sqrt{q_0-1/2}} + \frac{B_0}{t_1^{1/12}} \left(\frac{1}{6}\log t_1 + \log \tau_1 - \log(q_0-1/2)\right).$$

Choosing $t_1 = \exp(16)$, $\eta = 1.267$ and $q_0 = 5$, we obtain

$$|\zeta(1+it)| \leq \frac{1}{3}\log t + 4.66, \qquad t \geq \exp(16). \tag{61}$$

This implies Theorem 1 in the range $\exp(16) \leq t \leq \exp(88)$.

Consider now the second part of Proposition 3. In this critical region, we make use of a few additional tools to sharpen our estimates. In particular, we employ a different method of bounding the second sum appearing in Theorem 9, leading to an estimate of size $O(t^{-1/12})$ instead of $O(1)$. Also, we employ the k-th derivative tests used in the proof of Theorem 1, however this time we introduce additional scaling parameters to further optimise the switching points between derivative tests of different order.

To this end, let t_2 be a real number to be chosen later, subject to $t_2 \geq \exp(182/3)$. Assume $t \geq t_2$. Recall from (58) and (59) we showed for $q_0 \geq 1$, $P = \lceil t^{1/3} \rceil$ and $t \geq t_2$ that the tail sum satisfies

$$\left|\sum_{n=q_0P}^{n_1} \frac{1}{n^{1+it}}\right| \leq \frac{\alpha_0^{1/2}}{\pi^{1/6}} \frac{2}{\sqrt{q_0-1/2}} + \frac{B_0}{t_2^{1/12}} \left(\frac{1}{6}\log t_2 + \log \tau_1 - \log(q_0-1/2)\right). \tag{62}$$

As for the initial sum over $1 \leq n < q_0P$, instead of bounding it using the triangle inequality throughout as in (54), we will refine our estimate by using k-th order van der Corput lemmas with $k = 4$ and 5 in the subintervals $(Z_5, Z_4]$ and $(Z_6, Z_5]$ respectively, where

$$Z_4 := q_0P - 1, \qquad Z_5 := \mu_5 t^{\theta_5/2}, \qquad Z_6 := \mu_6 t^{\theta_6/2},$$

with θ_5, θ_6 the same as in the proof of Theorem 1 and μ_5, μ_6 real positive numbers to be chosen later, subject to

$$1 < \mu_6 t_2^{\theta_6/2} \le \mu_5 t_2^{\theta_5/2} \le q_0 \lceil t_2^{1/3} \rceil - 1. \tag{63}$$

We remark that Z_5, Z_6 will serve a similar purpose as Y_k in the proof of Theorem 1, whereas the new parameters μ_5, μ_6 will allow us to fine-tune our estimates in this critical region.

Let $h' > 1$ be a real number to be chosen later, and let $X'_m = (h')^{-m} Z_4$ for integer m. Define

$$M' := \left\lceil \frac{\log(Z_4/Z_6)}{\log h'} \right\rceil.$$

The definition of M' guarantees $X'_{M'} \le Z_6 < X'_{M'-1}$. Next, define the subsum

$$S'_m := \sum_{X'_m + 1 < n \le X'_{m-1}} \frac{1}{n^{1+it}},$$

so that, similarly to (12),

$$\left| \sum_{X'_{M'-1} < n \le Z_4} \frac{1}{n^{1+it}} \right| \le \sum_{1 \le m \le M'-1} \left(\frac{1}{X'_m} + |S'_m| \right).$$

By a similar calculation as in (13), for each $1 \le m \le M' - 1$,

$$\frac{\lfloor X'_{m-1} \rfloor}{\lfloor X'_m \rfloor + 1} \le h'.$$

Let $k = 4$ or 5. Following the argument of Theorem 1, if $Z_{k+1} < X'_m \le Z_k$, then

$$\begin{aligned} |S'_m| &\le C_k(\eta, h')(\lfloor X'_m \rfloor + 1)^{-k/(2K-2)} t^{1/(2K-2)} \\ &\qquad + D_k(\eta, h')(\lfloor X'_m \rfloor + 1)^{k/(2K-2) - 2/K} t^{-1/(2K-2)} \\ &\le C_k(\eta, h') Z_{k+1}^{-k/(2K-2)} t^{1/(2K-2)} + D_k(\eta, h')(Z_k + 1)^{k/(2K-2) - 2/K} t^{-1/(2K-2)}. \end{aligned}$$

We write $\mu_4 := q_0 + (q_0 - 1) t_2^{-1/3}$ so that $Z_4 \le \mu_4 t^{1/3}$. Since $k/(2K-2) - 2/K \ge 0$, for $t \ge t_2$,

$$|S'_m| \le A'_k(t) := \begin{cases} \mu_5^{-2/7} C_4 t^{-3/182} + (\mu_4 + t_2^{-1/3})^{1/28} D_4 t^{-5/84}, & k = 4, \\ \mu_6^{-1/6} C_5 t^{-7/990} + (\mu_5 + t_2^{-4/13})^{1/24} D_5 t^{-4/195}, & k = 5. \end{cases}$$

Let N'_k be the number of X'_m satisfying $Z_{k+1} < X'_m \le Z_k$. Then

$$N'_k \le \frac{\log Z_k - \log Z_{k+1}}{\log h'} + 1, \qquad \text{therefore} \qquad N'_k \le B'_k \frac{\log t}{\log h'},$$

where

$$B'_k := \begin{cases} \dfrac{1}{39} + \dfrac{\max\{0, \log \mu_4 - \log \mu_5 + \log h'\}}{\log t_2}, & k = 4, \\ \dfrac{28}{429} + \dfrac{\max\{0, \log \mu_5 - \log \mu_6 + \log h'\}}{\log t_2}, & k = 5. \end{cases}$$

Put together, we have

$$\sum_{Z_{k+1} < X'_m \le Z_k} |S'_m| \le \frac{A'_k(t) B'_k}{\log h'} \log t.$$

For $t \geq t_2$, the function $A_4'(t)\log t$ is decreasing, so $A_4'(t)\log t \leq A_4'(t_2)\log t_2$. For $k = 5$, we use

$$t^{-7/990}\log t \leq \frac{990}{7e} \qquad \text{and} \qquad t^{-4/195}\log t \leq t_2^{-4/195}\log t_2, \qquad (t \geq t_2).$$

Therefore,

$$\sum_{1\leq m\leq M'-1} |S_m'| \leq \frac{1}{\log h'}\bigg(A_4'(t_2)B_4'\log t_2 + \frac{990}{7e}\frac{B_5'C_5}{\mu_6^{1/6}} + (\mu_5 + t_2^{-4/13})^{1/24}B_5'D_5 t_2^{-4/195}\log t_2\bigg). \tag{64}$$

Next, similarly to (39), we obtain

$$\sum_{1\leq m\leq M'-1}\frac{1}{X_m'} < \frac{h'}{h'-1}\frac{1}{Z_6} \leq \frac{h'}{(h'-1)\mu_6 t_2^{\theta_6/2}}, \qquad (t \geq t_2). \tag{65}$$

In the remaining range $1 \leq n \leq X_{M'-1}'$, we use that $h'Z_6 \geq X_{M'-1}'$. Combined with the triangle inequality and (2), this yields

$$\begin{aligned}\left|\sum_{1\leq n\leq \lfloor X_{M'-1}\rfloor}\frac{1}{n^{1+it}}\right| &\leq \sum_{1\leq n\leq \lfloor h'Z_6\rfloor}\frac{1}{n} \leq \log h'Z_6 + \gamma + \frac{1}{2(h'Z_6-1)} \\ &\leq \frac{\theta_6}{2}\log t + \log\mu_6 + \log h' + \gamma + \frac{1}{2\big(h'\mu_6 t_2^{\theta_6/2}-1\big)}.\end{aligned} \tag{66}$$

Next, consider the second sum $\sum_{1\leq n\leq n_1} n^{it}$ in Theorem 9, which was just bounded by n_1 in our previous calculations. Let q, q_0, P, Q and N be as defined in the proof of Proposition 3, and suppose as before that $t \geq t_2 \geq \exp(182/3)$. Using $P \leq t^{1/3}+1$, the estimate (57) gives for $q_0 \leq q \leq Q$,

$$\max_{1\leq L\leq P}|S_N(L)| \leq (t^{1/3}+1)\left(\frac{\alpha_0^{1/2}}{\pi^{1/6}q^{1/2}} + \frac{B_0}{t^{1/12}}\right).$$

We combine this with the inequality

$$\sum_{q=q_0}^{Q}\frac{1}{q^{1/2}} \leq \int_{q_0-1}^{Q}\frac{\mathrm{d}x}{x^{1/2}} < 2Q^{1/2},$$

as well as bound the sum over $1 \leq n < q_0P$ by the number of terms in the sum, which gives

$$\left|\sum_{n=1}^{n_1} n^{it}\right| < q_0P - 1 + (t^{1/3}+1)\left[\frac{2\alpha_0^{1/2}Q^{1/2}}{\pi^{1/6}} + \frac{B_0Q}{t^{1/12}}\right].$$

Since $Q \leq t^{1/6}/\sqrt{2\pi}$, we obtain for $t \geq t_2$,

$$\left|\sum_{n=1}^{n_1} n^{it}\right| < C\,t^{5/12}, \tag{67}$$

where

$$C(q_0,\eta,t_2) := (1+t_2^{-1/3})\left[\frac{q_0}{t_2^{1/12}} + \frac{2^{3/4}\alpha_0^{1/2}}{\pi^{5/12}} + \frac{B_0}{\sqrt{2\pi}}\right].$$

Therefore, after multiplying by $g(t)t^{-1/2}$, we obtain a sharper bound on the contribution of the second sum in Theorem 9. Namely, the bound $Cg(t_2)\,t^{-1/12}$, which replaces the original estimate $g(t_2)/\sqrt{2\pi}$.

Lastly, combining (62), (64), (65), (66) and (67), and noting that $\theta_6 = 16/33$, we have for $t \geq t_2$

$$|\zeta(1+it)| \leq \frac{8}{33}\log t + B_3,$$

where

$$\begin{aligned} B_3 := \frac{1}{\log h'}\left(A_4'(t_2)B_4'\log t_2 + \frac{990}{7e}\frac{B_5'C_5}{\mu_6^{1/6}} + (\mu_5 + t_2^{-4/13})^{1/24}B_5'D_5t_2^{-4/195}\log t_2\right) \\ + \frac{h'}{(h'-1)\mu_6 t_2^{\theta_6/2}} + \frac{\alpha_0(W_0,\lambda_0)^{1/2}}{\pi^{1/6}}\frac{2}{\sqrt{q_0-1/2}} \\ + \frac{B_0(W_0,\lambda_0)}{t_2^{1/12}}\left(\frac{1}{6}\log t_2 + \log\tau_1 - \log(q_0-1/2)\right) \\ + \log\mu_6 + \log h' + \gamma + \frac{1}{2\big(h'\mu_6 t_2^{\theta_6/2}-1\big)} + C(q_0,\eta,t_2)\frac{g(t_2)}{t_2^{1/12}} + \mathcal{R}(t_2). \end{aligned}$$

We choose

$$\begin{aligned} \eta &= 1.3348, & h' &= 1.056, & q_0 &= 46, \\ \mu_5 &= 48.575, & \mu_6 &= 51.296, & t_2 &= \exp(90). \end{aligned}$$

Verifying that (63) holds, we calculate $B_3 \leq 12.45$, i.e. that

$$|\zeta(1+it)| \leq \frac{8}{33}\log t + 12.45, \qquad t \geq \exp(88).$$

This implies Theorem 1 for $\exp(88) \leq t \leq \exp(662)$.

6. Proof of Theorem 2

We begin by proving some bounds on $\zeta(s)$. To help make our notation more intuitive, we will allow reuse of variable names from previous sections. The reader should keep in mind these variables mean different things in the confines of this section. Since this section is independent from Section 3, Section 4 and Section 5, this will hopefully cause no confusion.

Lemma 4. *Let $C > 0$ and $t_0 \geq e^e$ be constants. Suppose $t \geq t_0$. Define*

$$\sigma_t := 1 - \frac{C(\log\log t)^2}{\log t}.$$

Additionally, suppose that C satisfies

$$\frac{C(\log\log t_e)^2}{\log t_e} \leq \frac{1}{2}, \qquad \text{where} \qquad t_e := e^{e^2}. \tag{68}$$

For each $t \geq t_0$, if $\sigma_t \leq \sigma \leq 2$, then

$$|\zeta(\sigma+it)| \leq A\log^B t \qquad \text{where} \qquad A = 76.2, \qquad B = \frac{2}{3} + \frac{71.2\,C^{3/2}}{e^2}. \tag{69}$$

Proof. From Ford [For02], if $1/2 \le \sigma \le 1$, then

$$|\zeta(\sigma+it)| \le 76.2\, t^{4.45(1-\sigma)^{3/2}} \log^{2/3} t. \tag{70}$$

As $\sigma_t \ge 1/2$, a consequence of the assumption (68) and considering that $(\log\log t)^2/\log t$ reaches its maximum at $t = t_e$, we may use (70) for any $\sigma \in [\sigma_t, 1]$.

In view of this, since $(1-\sigma)^{3/2}$ is monotonically decreasing with σ, if $\sigma_t \le \sigma \le 1$, then (70) gives

$$|\zeta(\sigma+it)| \le 76.2 \exp\left[4.45(1-\sigma_t)^{3/2}\log t + \frac{2}{3}\log\log t\right]. \tag{71}$$

Or, written differently,

$$|\zeta(\sigma+it)| \le A \exp\left(\tilde{B}\log\log t\right), \tag{72}$$

where A is defined as in (69) and

$$\tilde{B} := \frac{2}{3} + 4.45\, C^{3/2}\, \frac{(\log\log t)^2}{\sqrt{\log t}}.$$

As $(\log\log t)^2/\sqrt{\log t}$ reaches its maximum at $t = e^{e^4}$, attaining the value $16/e^2$, we obtain after a simple computation that $\tilde{B} \le B$, with B defined as in (69). The desired result therefore follows for $\sigma_t \le \sigma \le 1$ on replacing $\tilde{B}$ with B in (72).

To show that the result holds for $1 \le \sigma \le 2$ as well, we use the Phragmén–Lindelöf Principle on the holomorphic function

$$f(s) = (s-1)\zeta(s).$$

To this end, on the 1-line we have

$$|f(1+it)| = |t|\,|\zeta(1+it)| \le 62.6\,|2+it|\log^{2/3}|2+it|.$$

This inequality is verified numerically for $|t| < 3$ and is a consequence of the bound $|\zeta(1+it)| \le 62.6\log^{2/3} t$ for $|t| \ge 3$, given in [Tru14a]. Also, on the 2-line, plainly,

$$|f(2+it)| \le |1+it|\,\zeta(2) < 62.6|3+it|\log^{2/3}|3+it|,$$

for all real t. By [Tru14b, Lemma 3], we thus obtain[7]

$$|f(s)| \le 62.6|1+s|\log^{2/3}|1+s|, \qquad (1 \le \sigma \le 2). \tag{73}$$

Moreover, for $1 \le \sigma \le 2$ we have the easy estimates,

$$\frac{|1+s|}{|s-1|} \le \frac{\sqrt{9+t^2}}{t} \le \sqrt{1+9t_0^{-2}},$$

$$\log|1+s| \le \log\sqrt{9+t^2} \le \log t + \log\sqrt{1+9t_0^{-2}}.$$

So that, for $1 \le \sigma \le 2$ and $t \ge t_0$, the inequality (73) implies that

$$|\zeta(\sigma+it)| \le 62.6\sqrt{1+9t_0^{-2}}\left(1 + \frac{\log\sqrt{1+9t_0^{-2}}}{\log t_0}\right)\log^{2/3} t < A\log^B t,$$

where A and B are defined as in (69), as desired. □

[7]We apply [Tru14b, Lemma 3] with the following parameter values (in the notation in the cited paper): $a = 1$, $b = 2$, $Q = 1$, $\alpha_1 = \beta_1 = 1$, $\alpha_2 = \beta_2 = 2/3$ and $A = B = 62.6$. Note that the A and B in [Tru14b, Lemma 3] are different from the A and B in the statement of our lemma.

Remark. Following Titchmarsh's argument [Tit86, Theorem 5.17], we could make use of [Yan24, Theorem 1.1]. For simplicity, however, we used the Richert-type bound (70) due to Ford [For02] instead.

Lemma 5. *Let $t_0 \geq e^e$ and $d_1 > 0$ be constants. Suppose $t \geq t_0$. We have*

$$\left|\zeta\left(1 + d_1 \frac{\log\log t}{\log t} + it\right)\right| \leq C'(d_1, t_0) \frac{\log t}{\log\log t},$$

where γ is Euler's constant, and

$$C'(d_1, t_0) = \frac{1}{d_1} \exp\left(\gamma d_1 \frac{\log\log t_0}{\log t_0}\right). \tag{74}$$

Proof. We use the following uniform bound due to Ramaré [Ram16, Lemma 5.4]:

$$|\zeta(\sigma + it)| \leq \zeta(\sigma) \leq \frac{e^{\gamma(\sigma-1)}}{\sigma - 1}, \qquad (\sigma > 1).$$

Substituting $\sigma = 1 + d_1 \log\log t/\log t$ therefore gives for $t \geq t_0$,

$$\zeta(\sigma) \leq \frac{e^{\gamma d_1 \log\log t/\log t}}{d_1 \log\log t/\log t} \leq \frac{1}{d_1} \exp\left(\frac{\gamma d_1 \log\log t_0}{\log t_0}\right) \frac{\log t}{\log\log t},$$

as claimed. In the last inequality, we used that $\log\log t/\log t$ reaches its maximum at $t = e^e$ and is monotonically decreasing after that. □

6.1. **Bounding $|\zeta'(1+it)/\zeta(1+it)|$.** We follow the argument of [Tru15], which gives an explicit version of results in [Tit86, §3]. The method relies on Theorem 14.

We will therefore construct concentric disks, centred just to the right of the line $\sigma = 1 + it$, and extend their radius slightly to the left and into the critical strip. Ultimately, we want to apply Lemmas 15 and 16 with $f(s) = \zeta(s)$, which will give us the desired results.

The process is as follows. Let $t_0 \geq e^e$ and suppose $t' \geq t_0$. Let d be a real positive constant, which will be chosen later. Denote the center of the concentric disks to be constructed by $s' = \sigma' + it'$, where

$$\sigma' = 1 + \delta_{t'}, \qquad \text{with} \qquad \delta_{t'} := \frac{d \log\log t'}{\log t'}. \tag{75}$$

Notice that $\delta_{t'}$ is decreasing with t' and is at most d/e. Let $r > 0$ denote the radius of our largest concentric disk. In the sequel, r will be determined as function of s'.

In addition, suppose $\sigma' + r \leq 1 + \epsilon$ where $\epsilon \leq 1$ is another positive constant to be chosen later. This implies, for instance, that $r \leq \epsilon \leq 1$. In order to enforce this inequality on σ' and r, it is enough to demand that

$$\frac{d}{e} + r \leq \epsilon. \tag{76}$$

Let $s = \sigma + it$ be a complex number. Aiming to apply Lemma 15 in the disk $|s - s'| \leq r$, we first seek a valid A_1 in that disk. Similarly, for Lemma 16, we seek a valid A_2 at the disk center $s = s'$. In applying Lemma 16, we moreover need to ensure that the non-vanishing condition on $f(s) = \zeta(s)$ is fulfilled, which we do by way of a zero-free region of zeta.

Let us first determine a valid A_1, with the aid of Lemma 4. To this end, let $C > 0$ be a parameter restricted by the inequality in (68), so that (in the notation of Lemma 4) $\sigma_t \geq 1/2$ for $t \geq e$. Note that σ_t is monotonically decreasing for $e \leq t \leq t_e$ and monotonically increasing thereafter. So, the maximum of σ_t over

the interval $t'-r \le t \le t'+r$ occurs at one of the end-points. For example, if $t'-r \ge t_e$, then the maximum occurs at $t = t'+r$. And if $t'+r \le t_e$, then the maximum occurs at $t = t'-r$. With this in mind, we calculate

$$\max_{|t-t'|\le r} \sigma_t = \max\{\sigma_{t'-r}, \sigma_{t'+r}\} < 1 - \frac{C_0\,(\log\log t')^2}{\log t'}, \tag{77}$$

where, since $r \le \epsilon$,

$$C_0 := C\,\min\left\{\left(\frac{\log\log(t'-\epsilon)}{\log\log t'}\right)^2, \frac{\log t'}{\log(t'+\epsilon)}\right\}.$$

Clearly, $C_0 = C_0(t')$ increases monotonically with t' (towards C). We therefore set

$$C_1 := C_0(t_0) \qquad \text{and} \qquad \sigma_{1,t'} := 1 - \frac{C_1\,(\log\log t')^2}{\log t'}. \tag{78}$$

So that, by (77), no matter t', r and ϵ, so long as they are subject to our conditions,

$$\sigma_{1,t'} \ge \max_{|t-t'|\le r} \sigma_t.$$

Consequently, Lemma 4 gives that for each $t \in [t'-r, t'+r]$ and any $\sigma \in [\sigma_{1,t'}, 2]$, and on recalling $r \le \epsilon$,

$$|\zeta(s)| \le A\log^B(t'+r) \le A_3\log^B t', \qquad A_3 = A\left(1+\frac{\log(1+\epsilon/t')}{\log t'}\right)^B. \tag{79}$$

We would like the inequality (79) to hold throughout the disk $|s-s'| \le r$. This will be certainly the case if this disk lies entirely in the rectangle specified by $\sigma \in [\sigma_{1,t'}, 2]$ and $t \in [t'-r, t'+r]$. This follows, in turn, on requiring $\sigma' - r \ge \sigma_{1,t'}$ or, equivalently, that

$$r \le \frac{C_1(\log\log t')^2}{\log t'} + \delta_{t'}.$$

Therefore, subject to the constraint (76), the following choice of r works:

$$r = r_{t'} := \left(C_1 + \frac{d}{\log\log t'}\right)\frac{(\log\log t')^2}{\log t'}. \tag{80}$$

To ensure the constraint (76) is met for all $t' \ge t_0$, it suffices that

$$\frac{d}{e} + (C_1+d)\,\frac{4}{e^2} \le \epsilon \le 1. \tag{81}$$

Here, we used that $(\log\log t')^2/\log t'$ reaches its maximum of $4/e^2$ at $t' = t_e$.

Overall, combining (79) and (80) with the trivial bound $|\zeta(s')| \ge \zeta(2\sigma')/\zeta(\sigma')$, easily seen on considering the Euler product of zeta, we obtain that throughout the disk $|s-s'| \le r$,

$$\left|\frac{\zeta(s)}{\zeta(s')}\right| \le A_3\,\frac{\zeta(\sigma')}{\zeta(2\sigma')}\log^B t' = A_3\,\frac{X(\sigma')}{\sigma'-1}\log^B t',$$

where

$$X(\sigma') = \frac{\zeta(\sigma')(\sigma'-1)}{\zeta(2\sigma')}.$$

Since $\sigma' = \sigma'(t')$ decreases to 1 with $t' \geq t_0$, $X(\sigma')$ decreases to $1/\zeta(2)$ with t'. Thus, recalling from (75) that $\sigma' = 1 + \delta_{t'}$ and observing A_3 is also decreasing in $t' \geq t_0$, we obtain that throughout the disk $|s - s'| \leq r$,

$$\left|\frac{\zeta(s)}{\zeta(s')}\right| \leq \frac{A_{\max} X_{\max}}{d} \frac{\log^{B+1} t'}{\log\log t'}, \tag{82}$$

where $A_{\max} = A_3(t_0)$ and $X_{\max} = X(\sigma'(t_0))$.

In summary, when applying Lemma 15 in the disk $|s - s'| \leq r$, with r as in (80) and subject to (81), we may take

$$\log A_1 = (B+1)\log\log t' - \log\log\log t' + \log\left(\frac{A_{\max} X_{\max}}{d}\right).$$

Next, in preparation for applying Lemma 16, we want to ensure that the intersection of the disk $|s - s'| \leq r$ and the right half-plane $\sigma \geq \sigma' - \alpha r$ lies entirely in a zero-free region for zeta. By [Yan24, Corollary 1.2], we have the following zero-free region, valid for $t \geq 3$.

$$\zeta(\sigma + it) \neq 0 \quad \text{for} \quad \sigma > 1 - \frac{c_0 \log\log t}{\log t}, \quad \text{where} \quad c_0 = \frac{1}{21.233}.$$

Since $1 - c_0 \log\log t/\log t$ is monotonically increasing for $t \geq t_0$ and since $t' \geq t_0$ and $r \leq \epsilon$, it suffices for our purposes to require that

$$\sigma' - \alpha r \geq 1 - \frac{c_0 \log\log(t' + \epsilon)}{\log(t' + \epsilon)}.$$

In other words, it suffices that

$$\alpha \leq \frac{1}{r}\left[\delta_{t'} + \frac{c_0 \log\log(t' + \epsilon)}{\log(t' + \epsilon)}\right].$$

Therefore, subject to $\alpha < 1/2$ as demanded by Lemma 16, the following choice of α works.

$$\alpha = \frac{1}{r} \cdot \frac{(d + c_1)\log\log t'}{\log t'} = \frac{d + c_1}{d + C_1 \log\log t'}, \tag{83}$$

where (using monotonicity to deduce the inequality below)

$$c_1 := \frac{c_0 \log t_0}{\log(t_0 + \epsilon)}, \qquad \text{so that} \qquad c_1 \leq c_0. \tag{84}$$

In particular, the constraint $\alpha < 1/2$ is fulfilled if

$$\frac{d + c_1}{d + C_1 \log\log t_0} < \frac{1}{2}. \tag{85}$$

Now, to determine A_2, we utilise [Del87], which asserts that if $\sigma > 1$, then

$$\left|\frac{\zeta'(s)}{\zeta(s)}\right| \leq -\frac{\zeta'(\sigma)}{\zeta(\sigma)} < \frac{1}{\sigma - 1}.$$

Taking $s = s'$ in this inequality leads to

$$\left|\frac{\zeta'(s')}{\zeta(s')}\right| < \frac{1}{\delta_{t'}} = \frac{\log t'}{d \log\log t'}.$$

So, we may take A_2 in Lemma 16 to be

$$A_2 = \frac{\log t'}{d \log\log t'}.$$

Lastly, in applying Lemma 16, we need to be able to reach the 1-line from our position at s'. So, we set

$$\beta = \frac{\delta_{t'}}{\alpha r} = \frac{d}{c_1 + d}.$$

Clearly, $\beta < 1$, as required by Lemma 16. The conclusion of the lemma therefore holds throughout the disk $|s - s'| \le \alpha\beta r = \delta_{t'}$. This includes, in particular, the horizontal line segment $s = \sigma + it'$, with $1 \le \sigma \le 1 + 2\delta_{t'}$.

Putting it all together, and making the substitution $t' \to t$, Lemma 16 hence furnishes the bound

$$\left|\frac{\zeta'(s)}{\zeta(s)}\right| \le \frac{8\beta}{(1-\beta)(1-2\alpha)^2}\,\frac{\log A_1}{r} + \frac{1+\beta}{1-\beta}\,A_2, \tag{86}$$

valid in the region

$$1 \le \sigma \le 1 + \frac{2d\log\log t}{\log t}, \qquad t \ge t_0.$$

We now input the values we determined for A_1, A_2, r, α and β into (86). We also note

$$\frac{1+\beta}{1-\beta} = 1 + \frac{2d}{c_1}, \qquad \text{so that} \qquad \frac{1+\beta}{1-\beta}\,A_2 = \left(\frac{1}{d} + \frac{2}{c_1}\right)\frac{\log t}{\log\log t},$$

as well as

$$\frac{8\beta}{1-\beta} = \frac{8d}{c_1}, \qquad \frac{1}{(1-2\alpha)^2} = \left(\frac{C_1\log\log t + d}{C_1\log\log t - d - 2c_1}\right)^2.$$

Hence, on observing that $1/(1-2\alpha)^2$ is decreasing in t, we obtain the following lemma.

Lemma 6. *Let $t_0 \ge e^e$ and suppose that $t \ge t_0$. Let C, d and ϵ be any positive constants satisfying the constraints* (68), (81) *and* (85). *If*

$$1 \le \sigma \le 1 + \frac{2d\log\log t}{\log t},$$

then

$$\left|\frac{\zeta'(\sigma+it)}{\zeta(\sigma+it)}\right| \le Q_1(C,d,\epsilon,t_0)\frac{\log t}{\log\log t},$$

where

$$Q_1(C,d,\epsilon,t_0) := \lambda_1\left[B + 1 + \frac{1}{\log\log t_0}\log\left(\frac{A_{\max}X_{\max}}{d}\right)\right] + \lambda_2.$$

Here,

$$\lambda_1 := \frac{1}{C_1}\cdot\frac{8d}{c_1}\cdot\left(\frac{C_1\log\log t_0 + d}{C_1\log\log t_0 - d - 2c_1}\right)^2, \qquad \lambda_2 := \frac{1}{d} + \frac{2}{c_1}.$$

The number B is defined as in Lemma 4 and depends on C. The numbers $A_{\max}$ and $X_{\max}$ are defined as in (82) *and depend on ϵ and t_0. The number C_1 is defined in* (78), *and depends on C, ϵ and t_0. The number c_1 is defined in* (84) *and depends on ϵ, t_0 and $c_0 = 1/21.233$.*

6.2. **Bounding $|1/\zeta(1+it)|$.** Moving from a bound on the logarithmic derivative of ζ to one on the reciprocal of ζ is done in the usual way (see [Tit86, Theorem 3.11]), with some improvements using the trigonometric polynomial (see [Car+22, Proposition A.2]).

Lemma 7. *Let $t_0 \geq e^e$ and $d > 0$ be constants. Suppose that for each $t \geq t_0$,*

$$\text{if} \quad 1 \leq \sigma \leq 1 + \frac{2d \log\log t}{\log t}, \quad \text{then} \quad \left|\frac{\zeta'(\sigma+it)}{\zeta(\sigma+it)}\right| \leq R_1 \frac{\log t}{\log\log t}.$$

Then for any $t \geq t_0$ and any real number d_1 such that $0 < d_1 \leq 2d$, we have

$$\left|\frac{1}{\zeta(1+it)}\right| \leq Q_2(d_1, t_0) \frac{\log t}{\log\log t},$$

where

$$Q_2(d_1, t_0) = \exp(R_1 d_1) \left(\frac{\log\log t_0}{\log t_0} + \frac{1}{d_1}\right)^{3/4} \left(C'(d_1, 2t_0)\left(\frac{\log 2}{\log t_0} + 1\right)\right)^{1/4}.$$

The number C' is defined as in (74).

Proof. Suppose $t \geq t_0$. Let d_1 be a real positive parameter such that $d_1 \leq 2d$. Let $\delta_1 = d_1 \log\log t / \log t$. It is easy to see that

$$\begin{aligned}\log\left|\frac{1}{\zeta(1+it)}\right| &= -\mathrm{Re}\log\zeta(1+it) \\ &= -\mathrm{Re}\log\zeta\left(1+\delta_1+it\right) + \int_1^{1+\delta_1} \mathrm{Re}\left(\frac{\zeta'}{\zeta}(\sigma+it)\right)\mathrm{d}\sigma.\end{aligned}$$

So, using our suppositions, we have

$$\log\left|\frac{1}{\zeta(1+it)}\right| \leq -\log\left|\zeta\left(1+\delta_1+it\right)\right| + R_1 d_1. \tag{87}$$

On the other hand, for $\sigma > 1$, and using the classical nonnegativity argument involving the trigonometric polynomial $3+4\cos\theta+\cos 2\theta$ [Tit86, Section 3.3], gives that

$$\zeta^3(\sigma)|\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1.$$

Combining this with the inequality $\zeta(\sigma) \leq \sigma/(\sigma-1)$ and Lemma 5, we therefore obtain, on taking $\sigma = 1+\delta_1$,

$$\begin{aligned}\left|\frac{1}{\zeta(1+\delta_1+it)}\right| &\leq |\zeta(1+\delta_1)|^{3/4}|\zeta(1+\delta_1+2it)|^{1/4} \\ &\leq \left(\frac{1+\delta_1}{\delta_1}\right)^{3/4}\left(C'(d_1,2t_0)\frac{\log 2t}{\log\log 2t}\right)^{1/4}.\end{aligned} \tag{88}$$

We now observe

$$\begin{aligned}\frac{1+\delta_1}{\delta_1} &\leq \left(\frac{\log\log t_0}{\log t_0} + \frac{1}{d_1}\right)\frac{\log t}{\log\log t}, \\ \frac{\log 2t}{\log\log 2t} &\leq \left(\frac{\log 2}{\log t_0} + 1\right)\frac{\log t}{\log\log t}.\end{aligned} \tag{89}$$

So that, combining (89) and (88), and noting by (87) we have

$$\left|\frac{1}{\zeta(1+it)}\right| \leq \frac{\exp(R_1 d_1)}{|\zeta(1+\delta_1+it)|},$$

the lemma follows. □

6.3. **Numeric calculations.** We define the following upper bounds:

$$Q_1(C, d, \epsilon, t_0) \le R_1 \qquad \text{and} \qquad Q_2(d_1, t_0) \le R_2.$$

The numeric calculations for R_1 and R_2 proceed as follows. First fix $t_0 \ge e^e$ and a positive $\epsilon \le 1$. For the purposes of obtaining an upper bound R_1, we then optimise $Q_1(C, d, \epsilon, t_0)$ in Lemma 6 over C and d, all the while ensuring C, d, and ϵ are subject to our constraints (68), (81) and (85)

Thereafter, with these values of d and R_1, the relationship $d_1 \le 2d$ allows us to obtain R_2 by optimising $Q_2(d_1, t_0)$ in Lemma 7 over d_1.

Our choices of

$$t_0 = 500, \qquad \epsilon = 0.517,$$

$$C = \frac{e^2}{8}, \qquad d = 0.018, \qquad d_1 = 0.0065,$$

give us

$$r \le 0.502, \qquad \alpha \le 0.0381, \qquad \beta = 0.274\ldots,$$

and the upper bounds

$$R_1 = 154.5, \qquad R_2 = 431.7.$$

In summary, we have for $t \ge 500$,

$$\frac{1}{|\zeta(1+it)|} \le 431.7 \frac{\log t}{\log\log t}. \tag{90}$$

Furthermore, in the region

$$1 \le \sigma \le 1 + \frac{9}{250}\frac{\log\log t}{\log t},$$

we have for $t \ge 500$,

$$\left|\frac{\zeta'}{\zeta}(\sigma + it)\right| \le 154.5 \frac{\log t}{\log\log t}. \tag{91}$$

Finally, to cover the range from $3 \le t \le 500$ in (90), we refer to the proof of Proposition $A.2$ in [Car+22], where via interval arithmetic, they computed that for $2 \le t \le 500$,

$$\frac{1}{|\zeta(1+it)|} \le 2.079 \log t.$$

Comparing this with the estimate in (90), we see after a simple verification that the bound

$$\frac{1}{|\zeta(1+it)|} \le 431.7 \frac{\log t}{\log\log t}$$

holds for all $t \ge 3$.

This concludes our proof of Theorem 2. These bounds can certainly be improved. Some possible avenues of attack would be to treat the sum over zeros in the proof of Lemma 16 more carefully, or to use a larger zero-free region.

Remark. It is possible with our methods to directly get a bound like (91) for $t < 500$, especially if one were not concerned with the size of R_2. For instance, choosing $t_0 = 16$, $\epsilon = 0.508$, $C = e^2/8$, $d = 0.0147$ and $d_1 = 0.0056$, would return for $t \geq 16$ that if

$$1 \leq \sigma \leq 1 + \frac{147}{5000}\frac{\log\log t}{\log t},$$

then

$$\left|\frac{\zeta'}{\zeta}(\sigma + it)\right| \leq 177.5\frac{\log t}{\log\log t}.$$

While this would cause a modest jump in R_1, the size of R_2 would increase significantly to $R_2 = 511.1$. This is because R_2 increases exponentially with R_1, as evident from Lemma 7. The dominant factor there is C', and any improvement in that quantity would help greatly.

7. Background results

7.1. Background results related to Sections 3-5.

Lemma 8. *[Fra21, Lemma 2.1] Let γ denote Euler's constant. For $N \geq 1$,*

$$\sum_{n=1}^{N}\frac{1}{n} < \log N + \gamma + \frac{1}{2N} - \frac{1}{12N^2} + \frac{1}{64N^4}.$$

Theorem 9. *[Pat22, Theorem 3.9] For $t > 0$ and $n_1 = \lfloor\sqrt{t/(2\pi)}\rfloor$ we have*

$$|\zeta(1+it)| \leq \left|\sum_{n=1}^{n_1}\frac{1}{n^{1+it}}\right| + \frac{g(t)}{t^{1/2}}\left|\sum_{n=1}^{n_1}\frac{1}{n^{-it}}\right| + \mathcal{R}$$

where

$$g(t) = \sqrt{2\pi}\exp\left(\frac{5}{3t^2} + \frac{\pi}{6t}\right),$$

and

$$\mathcal{R} = \mathcal{R}(t) := \frac{1}{t^{1/2}}\left(\sqrt{\frac{\pi}{2}} + \frac{g(t)}{2}\right) + \frac{1}{t}\left(9\sqrt{\frac{\pi}{2}} + \frac{g(t)}{\sqrt{\pi(3-2\log 2)}}\right) + \frac{1}{t^{3/2}}\left(\frac{968\pi^{3/2} + g(t)242\pi}{700}\right).$$

Next we review some explicit estimates of exponential sums from the literature. The following lemma is a specialised third-derivative test for a phase function commonly encountered when estimating the Riemann zeta-function. It is due to [HPY24, Lemma 2.6] which builds on [Hia16].

Lemma 10 (Explicit third derivative test)**.** *[HPY24, Lemma 2.6] Let r and K be positive integers, and let t and K_0 be positive numbers. Suppose $K \geq K_0 > 1$. Let*

$$f(x) := \frac{t}{2\pi}\log(rK + x), \qquad W := \frac{\pi(r+1)^3K^3}{t}, \qquad \lambda := \frac{(1+r)^3}{r^3},$$

so that $1/W \leq |f'''(x)| \leq \lambda/W$ for $0 \leq x \leq K$. Then, for each positive integer $L \leq K$ and any $\eta > 0$,

$$\left|\sum_{n=0}^{L-1} e^{2\pi i f(n)}\right|^2 \leq \left(\frac{K}{W^{1/3}} + \eta\right)(\alpha K + \beta W^{2/3})$$

where

$$\alpha = \alpha(W,\lambda) := \frac{1}{\eta} + \frac{\eta\mu}{3W^{1/3}} + \frac{\mu}{3W^{2/3}} + \frac{32\mu}{15\sqrt{\pi}}\sqrt{\eta + W^{-1/3}},$$
$$\beta = \beta(W) := \frac{32}{3\sqrt{\pi\eta}} + \left(2 - \frac{4}{\pi}\right)\frac{1}{W^{1/3}},$$
$$\mu = \mu(\lambda) := \frac{1}{2}\lambda^{2/3}\left(1 + \frac{1}{(1-K_0^{-1})\lambda^{1/3}}\right).$$

Remark. If $W \le 1$, say, then α will be greater than 1, and so $\alpha L^2 \ge L^2$. Hence, for small W, it is better to use the triangle inequality than Lemma 10.

Remark. We mention that there is a typo in the statement of [HPY24, Lemma 2.6] where N in the statement of the lemma should be 0. This lemma is only applied with $N = 0$ in [HPY24] in any case.[8]

The next two lemmas respectively represent another third derivative test and a kth derivative test for $k \ge 4$, due to [Yan24]. The reason we state this additional third derivative test is because A_3 and B_3 comprise the boundary cases for the recursive formulas in Lemma 12.

Lemma 11 (Another explicit third derivative test)**.** *[Yan24, Lemma 2.4] Let a and N be integers such that $N \ge 1$. Suppose $f(x)$ has three continuous derivatives and $f'''(x)$ is monotonic over the interval $(a, a+N]$. Suppose further that there are numbers $\lambda_3 > 0$ and $\omega > 1$ such that $\lambda_3 \le |f'''(x)| \le \omega\lambda_3$ for all $x \in (a, a+N]$. Then for any $\eta > 0$,*

$$\left|\sum_{n=a+1}^{a+N} e^{2\pi i f(n)}\right| \le A_3\omega^{1/2}N\lambda_3^{1/6} + B_3N^{1/2}\lambda_3^{-1/6}$$

where

$$A_3 = A_3(\eta,\omega) := \sqrt{\frac{1}{\eta\omega} + \frac{32}{15\sqrt{\pi}}\sqrt{\eta + \lambda_0^{1/3}} + \frac{1}{3}\left(\eta + \lambda_0^{1/3}\right)\lambda_0^{1/3}}\delta_3,$$

$$B_3 = B_3(\eta) := \frac{\sqrt{32}}{\sqrt{3}\pi^{1/4}\eta^{1/4}}\delta_3,$$

and λ_0 and δ_3 are defined by

$$\lambda_0 = \lambda_0(\eta,\omega) := \left(\frac{1}{\eta} + \frac{32\eta^{1/2}\omega}{15\sqrt{\pi}}\right)^{-3},$$

$$\delta_3 = \delta_3(\eta) := \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{1 + \frac{3}{8}\pi^{1/2}\eta^{3/2}}}.$$

Lemma 12 (General explicit k-th derivative test)**.** *[Yan24, Lemma 2.5] Let a, N and k be integers such that $N \ge 1$ and $k \ge 4$. Suppose $f(x)$ is equipped with k continuous derivatives and $f^{(k)}(x)$is monotonic over the interval $(a, a+N]$. Suppose*

[8] In passing, let us also point out that on [HPY24, p. 211], the minimum in the definition of y_j should be against $L-m$ rather than $L-m-1$. This does not affect Equation 5.4 on p. 211 since, as stated in [HPY24, Lemma 2.5], the bound on $|g''(x)|$ holds up to $K-m$, not only $K-m-1$. None of the results in [HPY24] are affected.

further that there are numbers $\lambda_k > 0$ and $\omega > 1$ such that $\lambda_k \le |f^{(k)}(x)| \le \omega\lambda_k$ for all $x \in (a, a+N]$. Then for any $\eta > 0$,

$$\left|\sum_{n=a+1}^{a+N} e^{2\pi i f(n)}\right| \le A_k\omega^{2/K}N\lambda_k^{1/(2K-2)} + B_kN^{1-2/K}\lambda_k^{-1/(2K-2)},$$

where $K = 2^{k-1}$, A_k and B_k are defined recursively via the formulas

$$A_{k+1}(\eta,\omega) := \delta_k\left(\omega^{-1/K} + \frac{2^{19/12}(K-1)}{\sqrt{(2K-1)(4K-3)}}A_k(\eta,\omega)^{1/2}\right), \tag{92}$$

$$B_{k+1}(\eta) := \delta_k\frac{2^{3/2}(K-1)}{\sqrt{(2K-3)(4K-5)}}B_k(\eta)^{1/2}, \tag{93}$$

with A_3 and B_3 as in Lemma 11, and δ_k defined by

$$\delta_k = \delta_k(\eta) := \sqrt{1 + \frac{2}{2337^{1-2/K}}\left(\frac{9\pi}{1024}\eta\right)^{1/K}}. \tag{94}$$

Lemma 13. *Let a, b and k be integers such that $a > 0$, $b > a$ and $k \ge 3$. Let $t > 0$. Let $h > 1$ be a number such that $b \le ha$. Then for any $\eta > 0$,*

$$\left|\sum_{a<n\le b}\frac{1}{n^{1+it}}\right| \le C_ka^{-k/(2K-2)}t^{1/(2K-2)} + D_ka^{k/(2K-2)-2/K}t^{-1/(2K-2)},$$

where

$$C_k = C_k(\eta,h) := A_k(\eta,h^k)h^{2k/K-k/(2K-2)}(h-1)\left(\frac{(k-1)!}{2\pi}\right)^{\frac{1}{2K-2}},$$

$$D_k = D_k(\eta,h) := B_k(\eta)h^{k/(2K-2)}(h-1)^{1-2/K}\left(\frac{2\pi}{(k-1)!}\right)^{\frac{1}{2K-2}}.$$

and $A_k(\eta,h^k)$ and $B_k(\eta)$ defined in accordance with Lemma 12.

Proof. This is a special case of Lemma 12 and Lemma 11 with the phase function

$$f(x) = -\frac{t}{2\pi}\log x, \qquad \text{and hence} \qquad f^{(k)}(x) = (-1)^k\frac{t}{2\pi}\frac{(k-1)!}{x^k}.$$

Since $a < b \le ha$ by assumption, $\lambda_k \le |f^{(k)}(x)| \le \omega\lambda_k$ for $a < x \le b$, where

$$\lambda_k = \frac{t}{2\pi}\frac{(k-1)!}{h^ka^k}, \qquad \omega = h^k.$$

Suppose that $k \ge 4$. Then, we may apply Lemma 12 with these values of λ_k and ω, and with $N = b-a$. This furnishes a bound that increases with N. Noting that $N \le (h-1)a$, we thus obtain

$$\begin{aligned}\left|\sum_{a<n\le b}\frac{1}{n^{it}}\right| &\le A_k(\eta,h^k)(h^k)^{2/K}(h-1)a\left(\frac{t}{2\pi}\frac{(k-1)!}{h^ka^k}\right)^{1/(2K-2)}\\ &\qquad + B_k(\eta)((h-1)a)^{1-2/K}\left(\frac{t}{2\pi}\frac{(k-1)!}{h^ka^k}\right)^{-1/(2K-2)}\\ &= C_ka^{1-k/(2K-2)}t^{1/(2K-2)} + D_ka^{1-2/K+k/(2K-2)}t^{-1/(2K-2)}.\end{aligned}$$

The desired result hence follows from partial summation in this case. If $k = 3$, then the result follows from Lemma 11 using a similar calculation. □

7.2. **Background results related to Section 6.** Here we give some preliminary lemmas for our proofs of estimates on $|\zeta'(1+it)/\zeta(1+it)|$ and $|1/\zeta(1+it)|$. We reiterate that Section 6 reuses variable names from previous sections. These variables mean different things in the confines of this section as this section is independent from Sections 3, 4, and 5.

The first two lemmas, which form the basis of our method, are built upon a theorem of Borel and Carathéodory. This theorem enables us to deduce an upper bound for the modulus of a function and its derivatives on a circle, from bounds for its real part on a larger concentric circle.

Theorem 14 (Borel–Carathéodory)**.** *Let s_0 be a complex number. Let R be a positive number, possibly depending on s_0. Suppose that the function $f(s)$ is analytic in a region containing the disk $|s-s_0| \le R$. Let M denote the maximum of $\operatorname{Re} f(s)$ on the boundary $|s - s_0| = R$. Then, for any $r \in (0, R)$ and any s such that $|s - s_0| \le r$,*

$$|f(s)| \le \frac{2r}{R-r} M + \frac{R+r}{R-r} |f(s_0)|.$$

If in addition $f(s_0) = 0$, then for any $r \in (0, R)$ and any s such that $|s - s_0| \le r$,

$$|f'(s)| \le \frac{2R}{(R-r)^2} M.$$

Proof. Proofs for estimating both $|f(s)|$ and $|f'(s)|$ are available in [Tit39, §5.5]. However, if we include the restriction $f(s_0) = 0$, we can refer to [MV07, Lemma 6.2], which provides alternative proofs, resulting in a sharper estimate for $|f'(s)|$. □

Lemma 15. *Let s_0 be a complex number. Let r and α be positive numbers (possibly depending on s_0) such that $\alpha < 1/2$. Suppose that the function $f(s)$ is analytic in a region containing the disk $|s - s_0| \le r$. Suppose further there is a number A_1 independent of s such that,*

$$\left| \frac{f(s)}{f(s_0)} \right| \le A_1, \qquad |s - s_0| \le r.$$

Then, for any s in the disk $|s - s_0| \le \alpha r$ we have

$$\left| \frac{f'(s)}{f(s)} - \sum_{\rho} \frac{1}{s-\rho} \right| \le \frac{4 \log A_1}{r(1-2\alpha)^2},$$

where ρ runs through the zeros of $f(s)$ in the disk $|s - s_0| \le \frac{1}{2}r$, counted with multiplicity.

Proof. We follow the proof in [Tit86, Section 3]. Let g be the function

$$g(s) = f(s) \prod_{\rho} \frac{1}{s-\rho},$$

where ρ in the (finite) product runs through the zeros of $f(s)$, counted with multiplicity, that satisfy $|\rho - s_0| \le \frac{1}{2}r$. Since by construction the poles and zeros cancel,

g is analytic in $|s-s_0| \le r$, and does not vanish in the disk $|s-s_0| \le \frac{1}{2}r$. Therefore, the function

$$h(s) = \log \frac{g(s)}{g(s_0)},$$

where the logarithm branch is determined by $h(s_0) = 0$, is analytic in some region containing the disk $|s - s_0| \le \frac{1}{2}r$.

Now, on the circle $|s-s_0| = r$, we have $|s-\rho| \ge \frac{1}{2}r \ge |s_0-\rho|$. So, on this circle,

$$\left|\frac{g(s)}{g(s_0)}\right| = \left|\frac{f(s)}{f(s_0)} \prod_\rho \left(\frac{s_0-\rho}{s-\rho}\right)\right| \le \left|\frac{f(s)}{f(s_0)}\right| \le A_1.$$

The inequality also holds in the interior of the circle, by the maximum modulus principle. Hence, $\operatorname{Re} h(s) \le \log A_1$ throughout the disk $|s - s_0| \le r$.

Given this, we apply Theorem 14 to $h(s)$ with the concentric circles centered at s_0 of radii αr and $\frac{1}{2}r$. This yields throughout the disk $|s - s_0| \le \alpha r$ that

$$|h'(s)| \le \frac{4 \log A_1}{r(1-2\alpha)^2}. \tag{95}$$

On the other hand, since $g(s_0)$ is constant, we have

$$|h'(s)| = \left|\frac{g'(s)}{g(s)}\right| = \left|\frac{f'(s)}{f(s)} - \sum_\rho \frac{1}{s-\rho}\right|.$$

Combining this formula with the inequality (95) completes the proof. □

Lemma 16. *Let s and s_0 be complex numbers with real parts σ and σ_0, respectively. Let r, α, β, A_1 and A_2 be positive numbers, possibly depending on s_0, such that $\alpha < 1/2$ and $\beta < 1$. Suppose that the function $f(s)$ satisfies the conditions of Lemma 15 with r, α and A_1, and that*

$$\left|\frac{f'(s_0)}{f(s_0)}\right| \le A_2.$$

Suppose, in addition, that $f(s) \ne 0$ for any s in both the disk $|s - s_0| \le r$ and the right half-plane $\sigma \ge \sigma_0 - \alpha r$. Then, for any s in the disk $|s - s_0| \le \alpha\beta r$,

$$\left|\frac{f'(s)}{f(s)}\right| \le \frac{8\beta \log A_1}{r(1-\beta)(1-2\alpha)^2} + \frac{1+\beta}{1-\beta} A_2.$$

Proof. Using the inequality $-|z| \le -\operatorname{Re} z \le |z|$, valid for any complex number z, together with Lemma 15 in $|s - s_0| \le \alpha r$, we have throughout this disk,

$$-\operatorname{Re} \frac{f'(s)}{f(s)} \le \frac{4 \log A_1}{r(1-2\alpha)^2} - \sum_\rho \operatorname{Re} \frac{1}{s-\rho}, \tag{96}$$

where ρ runs through the zeros of f (with multiplicity) satisfying $|\rho - s_0| \le \frac{1}{2}r$. By the assumption on the nonvanishing of f, each ρ in this sum satisfies $\operatorname{Re}\rho < \sigma_0 - \alpha r$. On the other hand, all s in the disk $|s - s_0| \le \alpha r$ clearly satisfy $\sigma \ge \sigma_0 - \alpha r$. Hence,

$$\sum_\rho \operatorname{Re} \frac{1}{s-\rho} = \sum_\rho \frac{\sigma - \operatorname{Re}\rho}{|s-\rho|^2} > 0,$$

throughout the disk $|s - s_0| \le \alpha r$. We may thus drop the sum in (96) and still obtain a valid upper bound, call it M_1.

Now, $-f'(s)/f(s)$ is analytic in the disk $|s - s_0| \le \alpha r$ since by assumption f does not vanish there. We can therefore apply Theorem 14 to $-f'(s)/f(s)$ using the two concentric circles $|s - s_0| \le \alpha\beta r$ and $|s - s_0| \le \alpha r$, and the upper bound $M_1 \ge M$ on the circle $|s - s_0| \le \alpha r$, which yields the result. □

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